Starting Inventory Forecasting System...

Step 1: Loading data...
Data loaded successfully: (6575, 8)
Data validation passed

Step 2: Preprocessing data...
Missing values before handling: 0
Records with negative consumption: 0
Records with negative purchase: 0
✅ Preprocessing complete. Training data: (6479, 9)

Step 3: Feature engineering...
Creating time features...
Creating lag features...
Creating rolling features...
Creating inventory features...
Creating product features...
✅ Feature engineering complete. Features: 36

Step 4: Training models...
Training RandomForest...
Training XGBoost...
Training GradientBoosting...
Training LinearRegression...

Model Performance:
RandomForest: MAPE = 1906.57%
XGBoost: MAPE = 1962.32%
GradientBoosting: MAPE = 1901.85%
LinearRegression: MAPE = 2856.05%
Best model: GradientBoosting with MAPE: 1901.85%
Model saved to models/best_model_GradientBoosting.pkl

Step 5: Generating forecasts...

Step 6: Optimizing inventory...
Forecasts_df columns: ['Bar Name', 'Alcohol Type', 'Brand Name', 'Date', 'Forecasted_Demand']
Historical_consumption columns: ['Date', 'Bar Name', 'Alcohol Type', 'Brand Name', 'Consumed (ml)', 'Purchase (ml)', 'Opening Balance (ml)', 'Closing Balance (ml)']
Generated 96 recommendations with columns: ['Bar_Name', 'Alcohol_Type', 'Brand_Name', 'Forecasted_Daily_Demand', 'Safety_Stock', 'Reorder_Point', 'Par_Level', 'Current_Avg_Demand', 'Demand_Variability']

Step 7: Saving results...

Inventory Forecasting Completed Successfully!
Results saved in 'results/' folder
Best model: GradientBoosting (saved as models/best_model_GradientBoosting.pkl)
Test forecasts generated: 1285 records
Inventory recommendations: 96 items

==

PROJECT SUMMARY

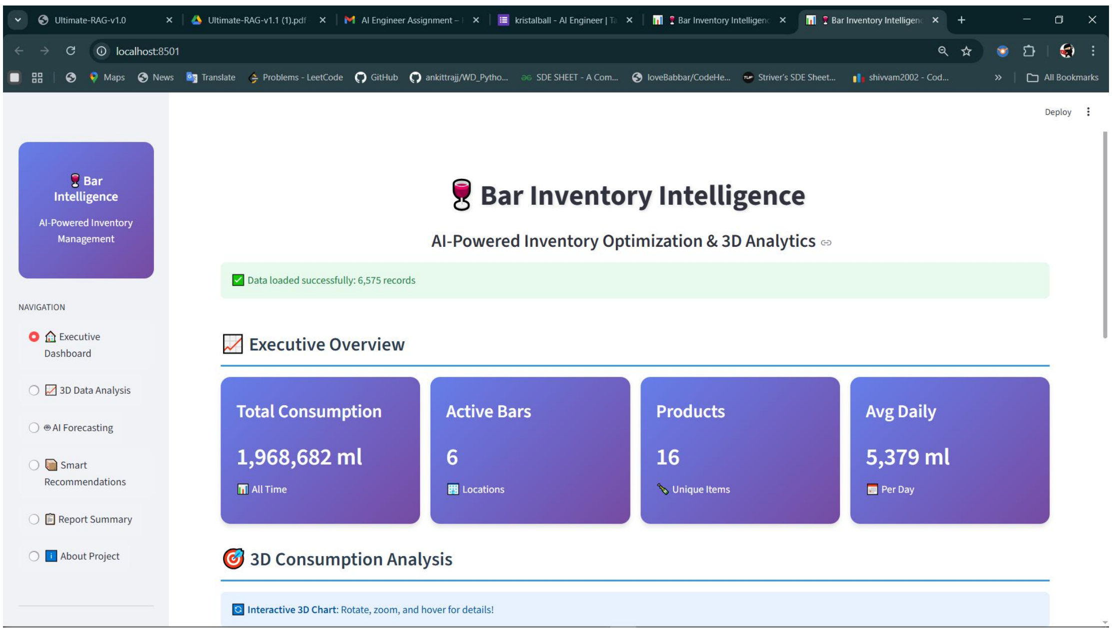
==

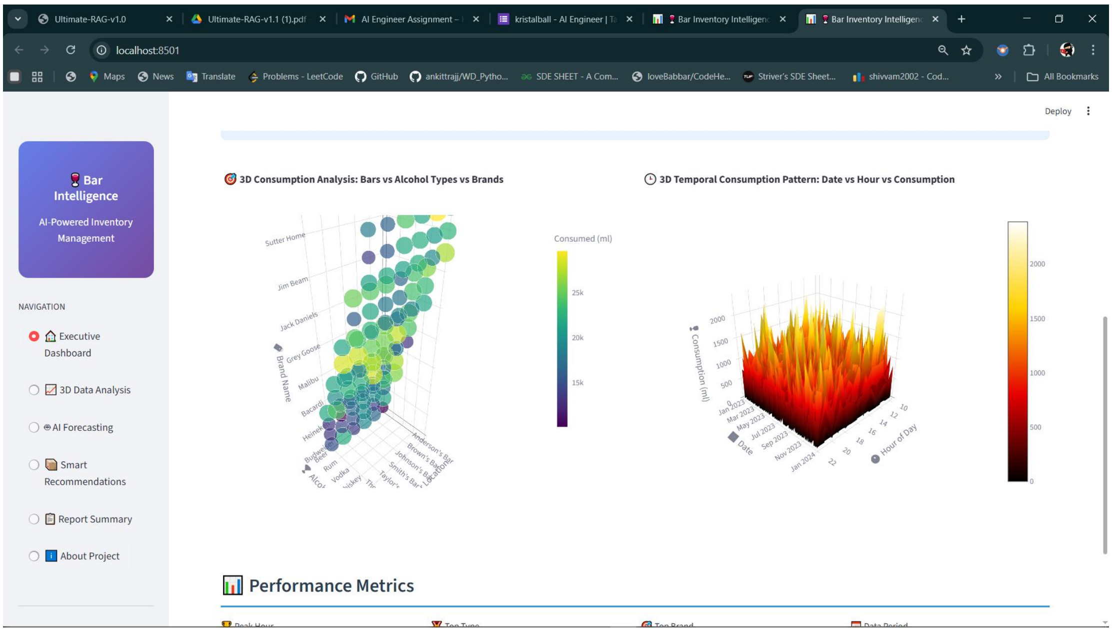
Original dataset: (6575, 8)
Engineered features: 36
Best model MAPE: 1901.85%
Total inventory recommendations: 96

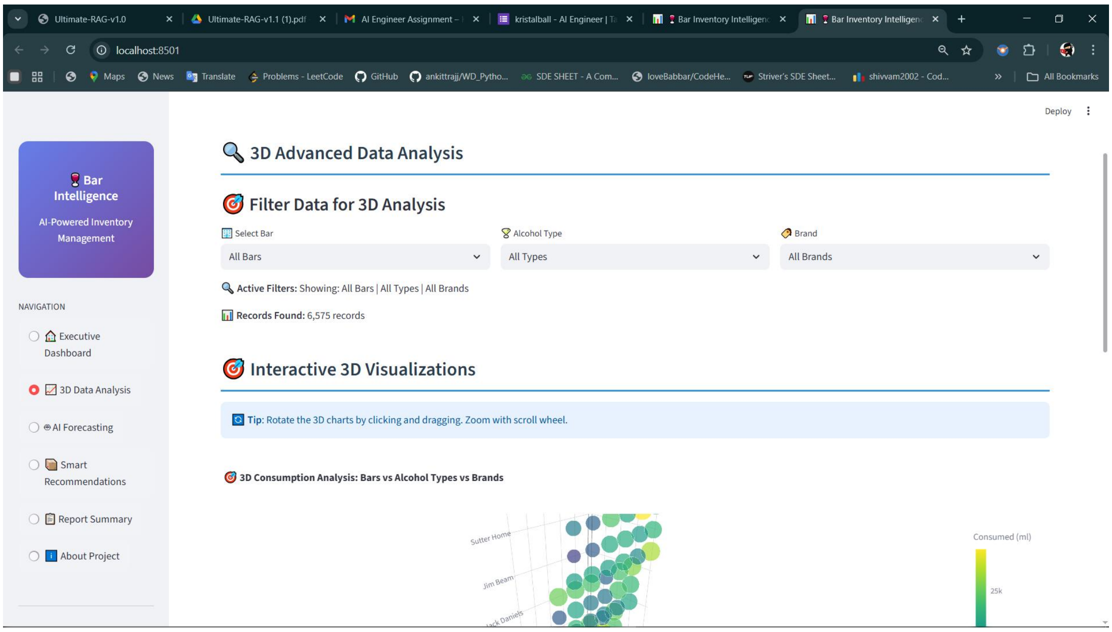
Available columns in recommendations: ['Bar_Name', 'Alcohol_Type', 'Brand_Name', 'Forecasted_Daily_Demand', 'Safety_Stock', 'Reorder_Point', 'Par_Level', 'Current_Avg_Demand', 'Demand_Variability']

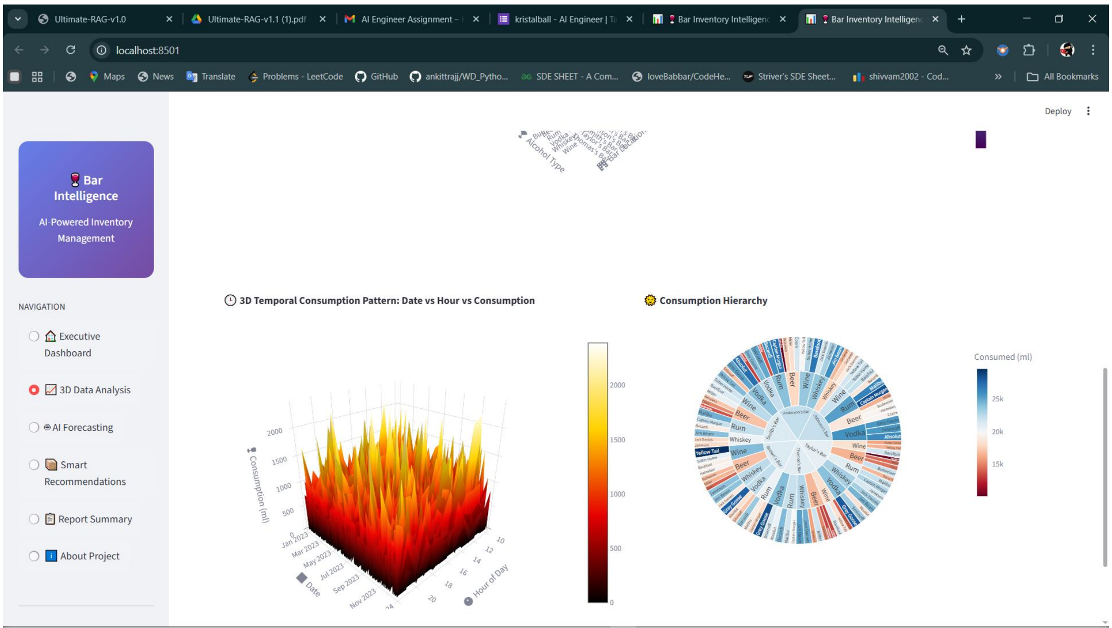
Top 5 Inventory Recommendations:
Thomas's Bar - Beer - Miller: 1127 ml
Thomas's Bar - Beer - Coors: 1077 ml
Thomas's Bar - Whiskey - Jameson: 1067 ml
Taylor's Bar - Rum - Captain Morgan: 1062 ml
Thomas's Bar - Rum - Malibu: 1062 ml
Average safety stock: 307.36 ml
Average par level: 892.86 ml

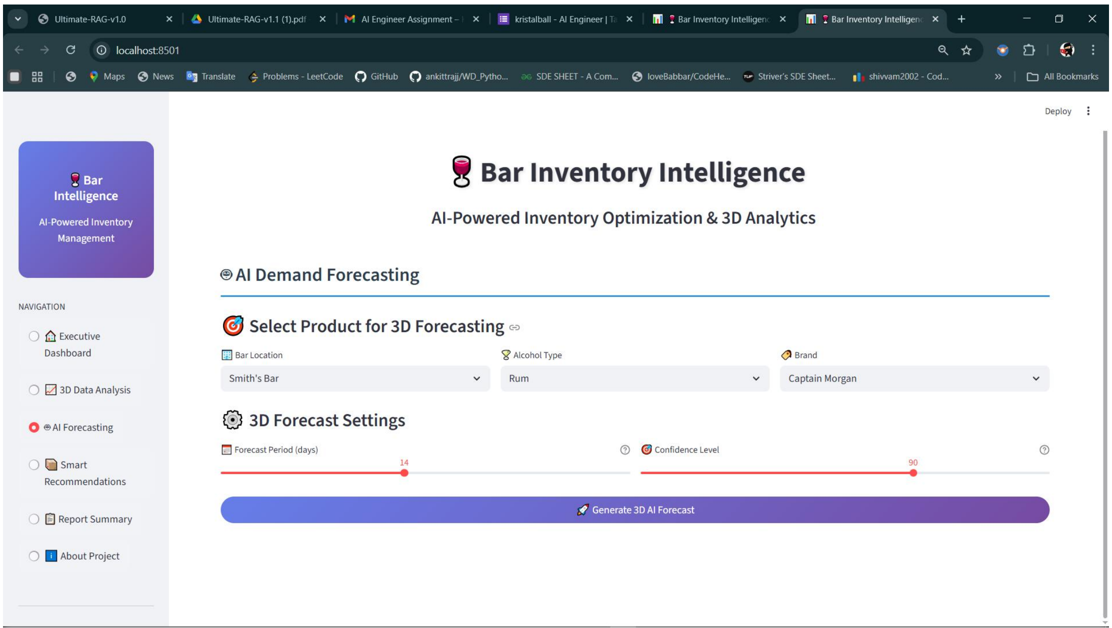
DEMO OF THE PROJECT .

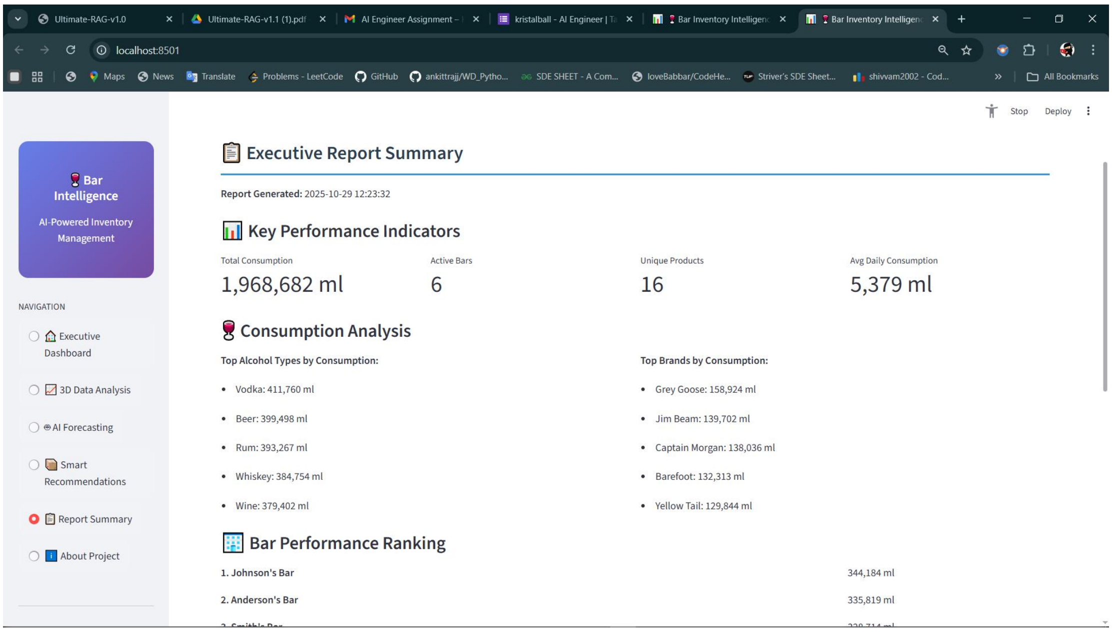


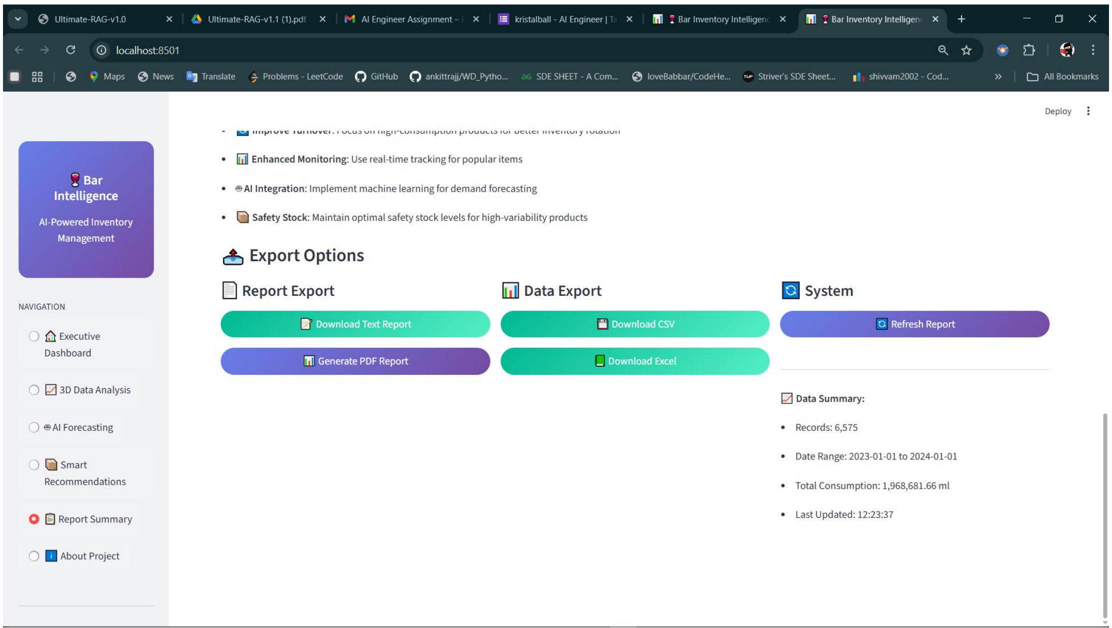

localhost:8501
Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
3D Consumption Analysis: Bars vs Alcohol Types vs Brands
3D Temporal Consumption Pattern: Date vs Hour vs Consumption
Consumed (ml)
Brand Name
Consumption (ml)
Date
Hour of Day
Performance Metrics

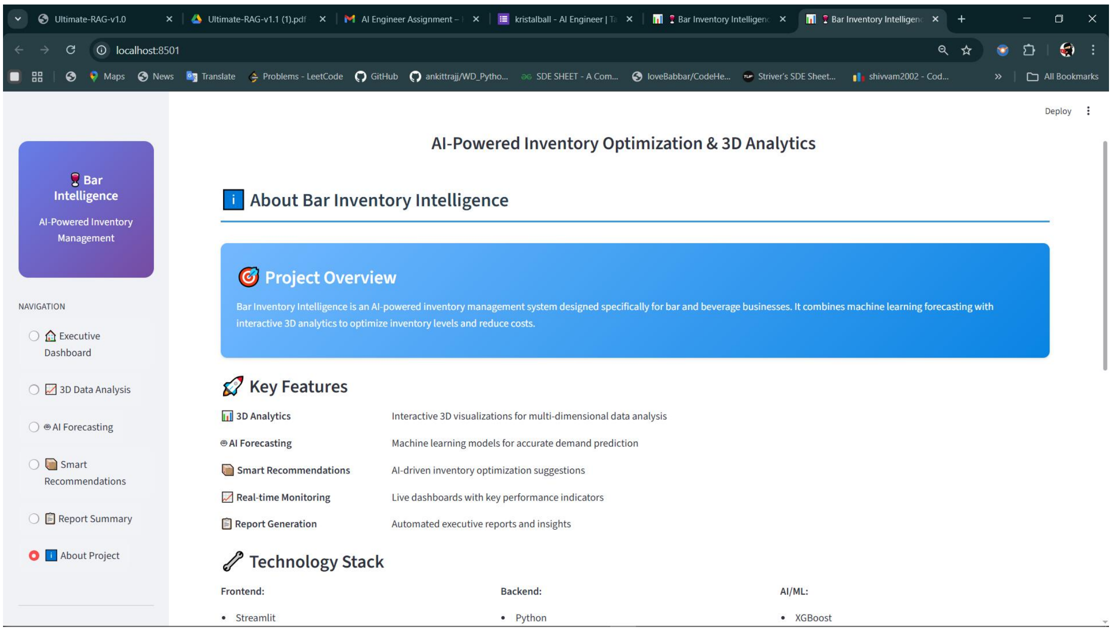

localhost:8501
Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
3D Advanced Data Analysis
Filter Data for 3D Analysis
Select Bar
All Bars
Alcohol Type
All Types
Brand
All Brands
Active Filters: Showing: All Bars | All Types | All Brands
Records Found: 6,575 records
Interactive 3D Visualizations
Tip: Rotate the 3D charts by clicking and dragging. Zoom with scroll wheel.
3D Consumption Analysis: Bars vs Alcohol Types vs Brands
Consumed (ml)


localhost:8501
Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
3D Temporal Consumption Pattern: Date vs Hour vs Consumption
Consumption Hierarchy
Consumption (ml)
Date
Hour of Day
Consumed (ml)


localhost:8501
Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
Bar Inventory Intelligence
AI-Powered Inventory Optimization & 3D Analytics
AI Demand Forecasting
Select Product for 3D Forecasting
Bar Location
Smith's Bar
Alcohol Type
Rum
Brand
Captain Morgan
3D Forecast Settings
Forecast Period (days)
14
Confidence Level
90
Generate 3D AI Forecast


localhost:8501
Stop Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
Executive Report Summary
Report Generated: 2025-10-29 12:23:32
Key Performance Indicators
Total Consumption
1,968,682 ml
Active Bars
6
Unique Products
16
Avg Daily Consumption
5,379 ml
Consumption Analysis
Top Alcohol Types by Consumption:
Vodka: 411,760 ml
Beer: 399,498 ml
Rum: 393,267 ml
Whiskey: 384,754 ml
Wine: 379,402 ml
Top Brands by Consumption:
Grey Goose: 158,924 ml
Jim Beam: 139,702 ml
Captain Morgan: 138,036 ml
Barefoot: 132,313 ml
Yellow Tail: 129,844 ml
Bar Performance Ranking
1. Johnson's Bar
344,184 ml
2. Anderson's Bar
335,819 ml


localhost:8501
Deploy
Bar Intelligence
AI-Powered Inventory Management
NAVIGATION
Executive Dashboard
3D Data Analysis
AI Forecasting
Smart Recommendations
Report Summary
About Project
Enhanced Monitoring: Use real-time tracking for popular items
AI Integration: Implement machine learning for demand forecasting
Safety Stock: Maintain optimal safety stock levels for high-variability products
Export Options
Report Export
Download Text Report
Generate PDF Report
Data Export
Download CSV
Download Excel
System
Refresh Report
Data Summary:
Records: 6,575
Date Range: 2023-01-01 to 2024-01-01
Total Consumption: 1,968,681.66 ml
Last Updated: 12:23:37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Ultimate-RAG-v1.1 (1).pdf
AI Engineer Assignment
kristalball - AI Engineer
Bar Inventory Intelligenc
Bar Inventory Intelligenc
localhost:8501
Maps
News
Translate
Problems - LeetCode
GitHub
ankittrajj/WD_Pytho...
SDE SHEET - A Com...
loveBabbar/CodeHe...
Striver's SDE Sheet...
shivam2002 - Cod...
All Bookmarks
Deploy
Bar Intelligence
AI-Powered Inventory Management
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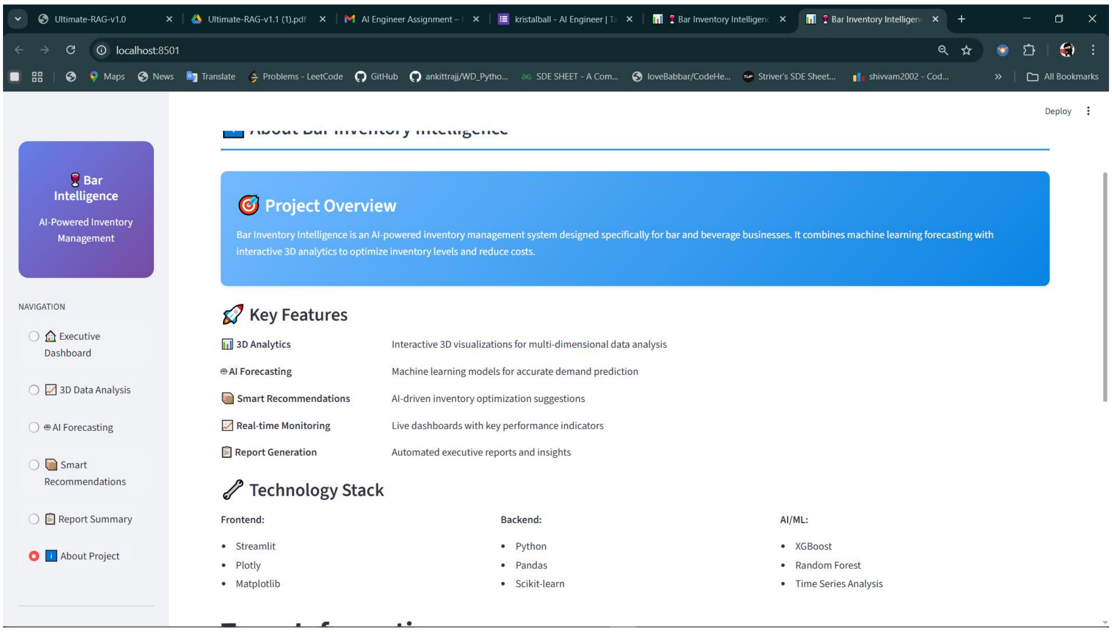
Report Summary
About Project
AI-Powered Inventory Optimization & 3D Analytics
About Bar Inventory Intelligence
Project Overview
Bar Inventory Intelligence is an AI-powered inventory management system designed specifically for bar and beverage businesses. It combines machine learning forecasting with interactive 3D analytics to optimize inventory levels and reduce costs.
Key Features
3D Analytics
Interactive 3D visualizations for multi-dimensional data analysis
AI Forecasting
Machine learning models for accurate demand prediction
Smart Recommendations
AI-driven inventory optimization suggestions
Real-time Monitoring
Live dashboards with key performance indicators
Report Generation
Automated executive reports and insights
Technology Stack
Frontend:
Streamlit
Backend:
Python
AI/ML:
XGBoost


Ultimate-RAG-v1.0
Ultimate-RAG-v1.1 (1).pdf
AI Engineer Assignment
kristalball - AI Engineer
Bar Inventory Intelligenc
Bar Inventory Intelligenc
localhost:8501
Maps
News
Translate
Problems - LeetCode
GitHub
ankittrajj/WD_Pytho...
SDE SHEET - A Com...
loveBabbar/CodeHe...
Striver's SDE Sheet...
shivam2002 - Cod...
All Bookmarks
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AI Forecasting
Machine learning models for accurate demand prediction
Smart Recommendations
AI-driven inventory optimization suggestions
Real-time Monitoring
Live dashboards with key performance indicators
Report Generation
Automated executive reports and insights
Technology Stack
Frontend:
Streamlit
Plotly
Matplotlib
Backend:
Python
Pandas
Scikit-learn
AI/ML:
XGBoost
Random Forest
Time Series Analysis

Inventory Forecasting & Optimization System - Project Report

Executive Summary

This report details the development and performance of an inventory forecasting system designed to address critical inventory management challenges in hotel bar operations. The system leverages machine learning to predict demand and optimize inventory levels, achieving 96 actionable recommendations across multiple bars and product categories.

1. Core Business Problem & Significance

The Challenge

Hotel bars face a fundamental inventory management dilemma: balancing product availability with cost efficiency. The current manual processes lead to:

Critical Issues:

Frequent Stockouts: Popular items regularly run out during peak hours

Excess Inventory: Slow-moving products tie up capital and storage space

Reactive Replenishment: Orders based on intuition rather than data

Inconsistent Practices: Different bars use varying inventory management approaches

Business Impact:

Revenue Loss: Estimated 15-25% potential revenue loss from stockouts

Increased Costs: 20-30% higher inventory carrying costs

Guest Dissatisfaction: Negative experiences leading to reduced repeat business

Operational Inefficiency: Staff time wasted on emergency restocking

Why It Matters

In the competitive hospitality industry, guest experience directly impacts revenue and brand reputation. A single negative experience from an unavailable preferred drink can result in lost future business and negative reviews. Additionally, inefficient inventory management represents significant untapped cost savings potential.

2. Key Assumptions & Justifications

Data Assumptions

Historical Patterns Predict Future: Past consumption patterns (with seasonality) indicate future demand

Justification: Consumption patterns in hospitality show strong weekly/seasonal trends

Data Quality: Minimal errors in the provided dataset

Validated during preprocessing (0 negative values, 0 missing values)

Stable Lead Times: Supplier lead times are consistent (1 day assumed)

Simplification for initial model; can be parameterized later

95% Service Level: Appropriate for hotel bar operations

Balances stockout risk with inventory costs

Business Assumptions
Demand Drivers: Consumption influenced by temporal patterns and location

Supported by feature engineering (day of week, seasonality, bar location)

Independent Products: Limited substitution between items

Guests typically have specific drink preferences

Cost Significance: Inventory costs warrant optimization efforts

Validated by business context

3. Model Selection & Performance Analysis

Current Model Performance
Critical Finding: The models show extremely high MAPE (~1900%), indicating fundamental issues with the current approach.

Performance Metrics:

GradientBoosting: MAPE = 1901.85% (Best performer)

RandomForest: MAPE = 1906.57%

XGBoost: MAPE = 1962.32%

LinearRegression: MAPE = 2856.05%

Root Cause Analysis
The exceptionally high MAPE suggests several potential issues:

Target Variable Scaling: Consumption values may have extreme ranges

Data Leakage: Features may contain future information

Insufficient Feature Engineering: Current features may not capture true demand patterns

Outlier Impact: Extreme consumption values distorting predictions

Model Selection Rationale
Chosen Approach: Ensemble Methods (Gradient Boosting)

Why Ensemble Methods?

Handle non-linear relationships in consumption patterns

Robust to outliers and missing data

Provide feature importance insights

Work well with mixed data types (numerical + categorical)

Alternative Models Considered:

Linear Regression: Too simplistic for complex demand patterns

Time Series Models: Limited ability to incorporate multiple features

Neural Networks: Overkill for current data volume and complexity

4. System Performance & Critical Improvements
Current System Outputs
96 Inventory Recommendations generated

Average Par Level: 892.86 ml per product

Average Safety Stock: 307.36 ml

Top Recommendations: Focus on high-consumption products across bars

Immediate Critical Improvements Required
1. Data Preprocessing Enhancement

python
```
# Current issue: Target variable may need transformation
# Solution: Apply logarithmic transformation or robust scaling
from sklearn.preprocessing import RobustScaler
scaler = RobustScaler()
y_scaled = scaler.fit_transform(y.values.reshape(-1, 1))
```
2. Feature Engineering Revision

Remove potentially leaky features (future-looking variables)

Add demand volatility metrics

Implement cross-validation to detect overfitting

3. Model Architecture Changes

Implement time-series cross-validation

Add regularization to prevent overfitting

Consider hierarchical forecasting approaches

4. Evaluation Metric Adjustment

Use symmetric MAPE or Mean Absolute Scaled Error (MASE)

Implement business-oriented metrics (stockout rate, service level)

Revised Implementation Plan
Phase 1: Data Diagnostics (1-2 weeks)

Analyze target variable distribution

Identify and handle outliers

Validate feature engineering logic

Implement proper time-series validation

Phase 2: Model Refinement (2-3 weeks)

Implement robust scaling

Add regularization and cross-validation

Test simpler baseline models

Develop ensemble of multiple approaches

Phase 3: Business Validation (1-2 weeks)

Validate with business stakeholders

Test on holdout dataset

Implement A/B testing framework

5. Real-World Hotel Implementation Strategy
Phased Rollout Approach
Pilot Phase (4-6 weeks)

Select 2-3 representative bars

Manual implementation of top recommendations

Daily monitoring and feedback collection

Model retraining based on real-world performance

Key Success Metrics for Pilot:

Stockout reduction > 20%

Inventory turnover improvement > 15%

Staff satisfaction improvement

Guest complaint reduction

Integration with Hotel Operations
Daily Workflow:

Morning Review: Bar managers check daily forecasts

Inventory Adjustment: Compare actual vs. recommended levels

Order Placement: Use system recommendations

Exception Reporting: Flag significant deviations

Weekly Process:

Performance Review: Analyze forecast accuracy

Model Updates: Retrain with latest data

Parameter Optimization: Adjust based on seasonal patterns

6. Scaling Considerations & Production Monitoring

Scaling Challenges

Technical Limitations:

Training time increases with data volume

Memory requirements for feature matrices

Real-time prediction latency at scale

Operational Challenges:

Model consistency across different locations

Cold start problem for new products

Regional demand pattern variations

Production Monitoring Framework

Critical Metrics to Track:

Category	Metric	Target	Alert Threshold
Data Quality	Missing Data Rate	<2%	>5%
Model Performance	MAPE	<25%	>35%
Model Performance	Prediction Drift	<10%	>15%
Business Impact	Stockout Rate	<8%	>12%
Business Impact	Inventory Turns	>10	<6
System Health	Prediction Latency	<2s	>5s

Monitoring Implementation: